

AirSense Cameroon

IndabaX 2026 Hackathon

Comparative Model Report & Reproducibility Guide

AI for Climate and Health Resilience in Cameroon

PM2.5 Prediction · 7-Day Forecasting · Health Alerts · 71 Cities

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1. Executive Summary

AirSense Cameroon is an end-to-end air quality prediction and health advisory system built for the IndabaX 2026 Hackathon. It uses six years of satellite reanalysis data (CAMS/ECMWF) combined with real-time meteorological data (Open-Meteo) to predict daily PM2.5 concentrations across 71 cities in all 10 regions of Cameroon.

The system moves beyond the starter-notebook baseline (which targets a synthetic proxy formula) to train directly on real CAMS satellite measurements, a genuinely harder and more meaningful benchmark. The deployment model achieves MAE = 6.27 $\mu\text{g}/\text{m}^3$ and $R^2 = 0.603$ on real CAMS data, compared to MAE = 9.05 and $R^2 = 0.208$ for the starter baseline, a 31% error reduction.

A full ML pipeline was implemented and evaluated: XGBoost variants, Ridge regression, LightGBM, a Transformer (Day+1/+2/+3), GraphSAGE GNN for spatial interpolation, and a Platt-calibrated alert classifier (AUC = 0.921, F1 = 0.847). The project also includes a FastAPI backend with 20+ REST endpoints, a Streamlit dashboard with six pages, conformal prediction intervals, CMIP6 climate projections to 2050, and a bilingual Groq-powered conversational AI assistant.

Metric	Value	Benchmark
Deployment Model MAE	6.27 $\mu\text{g}/\text{m}^3$	Starter: 9.05 $\mu\text{g}/\text{m}^3$
R^2 (real CAMS data)	0.603	Starter: 0.208
Alert Classifier AUC	0.921	Rule-based: F1=0.905
Alert Classifier F1	0.847 @ P=0.50	Platt calibrated
Conformal CI Coverage	97.7% ($\pm 17.34 \mu\text{g}/\text{m}^3$)	Target: 90%
Cities / Regions	71 / 10	Official dataset: 40
WHO Exceedance Rate	51.9% of city-days	Public health crisis

2. Project Architecture

The system follows a clean three-tier architecture: a data layer (CAMS + Open-Meteo + feature engineering), a model layer (XGBoost + ensemble + Transformer + GNN), and a presentation layer (FastAPI backend + Streamlit dashboard).

2.1 Repository Structure

The codebase is organised into six top-level directories:

Directory	Purpose
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dashboard/	Streamlit web application: 6 pages, sidebar nav, EN/FR, light/dark themes
api/	FastAPI backend with 20+ REST endpoints (forecast, alerts, SHAP, CMIP6, IoT ingest)
models/	Trained model artefacts: XGBoost (.json), calibrators, encoders, SHAP values
notebooks/	4 ordered notebooks: cleaning → EDA → modelling → pitch charts + parquet data files
scripts/	setup_models.py for rebuilding all artefacts from scratch without re-training
deployment/	render.yaml, the one-file Render.com deployment config for dashboard and API

2.2 Data Flow

Data moves through three stages: (1) ingestion from CAMS Reanalysis and Open-Meteo APIs, (2) feature engineering producing 68 model features, and (3) inference through XGBoost returning a PM2.5 prediction with conformal uncertainty bounds and a Platt-calibrated alert probability.

API-first design: the dashboard always attempts to route predictions through the FastAPI backend. If the backend is unreachable, it silently falls back to loading xgb_pm25.json locally with no visible difference to the user. This pattern is implemented in dashboard/utils/models.py.

3. Data Pipeline

3.1 Data Sources

Source	Period	Resolution	Variables
CAMS Reanalysis (Copernicus/ECMWF)	2022–2025	Daily, city-level	PM2.5, PM10, dust, CO, NO ₂ , O ₃ , SO ₂ , AOD
Open-Meteo ERA5 Archive	2020–2025	Daily, city-level	Temp, humidity, wind, precip, BLH, VPD, evapotranspiration, solar
Official hackathon dataset	2020–2024	Daily, 40 cities	26 meteorological variables (base dataset)
Proxy targets (engineered)	2020–2022	Daily	Pre-CAMS PM2.5 estimates validated against CAMS overlap period

3.2 Dataset Statistics (Final)

The fully engineered dataset contains 87,240 rows (one row per city-day) and 80 columns with zero null values. It spans 40 cities across all 10 regions from 2020-01-01 to 2025-12-20. Of the total rows,

56.6% use real CAMS satellite measurements and 43.4% use validated proxy targets for the pre-CAMS period.

A critical finding during EDA: 51.9% of all city-days exceed the WHO 24-hour PM2.5 limit of 15 µg/m³, representing a genuine public health emergency. The year 2024 was identified as the worst year on record, with national mean PM2.5 rising approximately 35% since 2020.

3.3 Feature Engineering

The model uses 68 engineered features. Key groups include:

Feature / Group	Description & Justification
nh3_proxy	Engineered biomass burning index; strongest correlate with PM2.5 at $r = +0.572$, outperforming every raw meteorological variable
vpd_x_dry_season	Top SHAP driver globally; product of vapour pressure deficit and dry-season flag; captures atmospheric aridity effect on PM2.5 accumulation
is_harmattan	Nov–Feb, northern region, and NE wind bearing combined to identify Saharan dust intrusion events (9.8% of days, 1.7× PM2.5 multiplier)
is_dry_season	Month in {11, 12, 1, 2, 3}; captures the seasonal pollution regime across all regions
is_dust_event	CAMS dust PM2.5 > 20 µg/m³; binary flag for Saharan dust episodes
month_sin / month_cos	Cyclical month encoding that preserves the circular nature of seasonal patterns
blh_min, blh_mean	Boundary Layer Height; lower BLH traps pollutants near the surface ($r = -0.099$ after BLH enrichment)
city_enc, region_enc	LabelEncoder-encoded spatial identifiers that allow the model to learn city and region-specific pollution baselines

4. EDA Findings & Modelling Decisions

Before any model was trained, we conducted a structured 18-section EDA on the full 87,240-row dataset. Every decision in the modelling notebook, from log-transforms to feature inclusion to sample weighting, traces back to a specific EDA finding. This section documents what we found, and what we decided as a result.

4.1 Target Variable Distribution

The first question was: what does the target variable look like?

Compound	Mean ($\mu\text{g}/\text{m}^3$)	Median	Max	WHO Exceedance
PM2.5 (primary target)	18.86	15.47	335.2	51.9% of city-days
PM10	34.82	24.43	670.9	24.9%
Dust	31.47	10.71	600.0	Binary character: 90% near-zero, spikes to 600 during Harmattan
CO, NO ₂ , SO ₂	n/a	n/a	n/a	0.0% (well within WHO limits)

PM2.5 is strongly right-skewed (skewness ≈ 2.37). Most city-days are moderate, but a small number of extreme Harmattan episodes push the distribution to 335 $\mu\text{g}/\text{m}^3$. Crucially, the median PM2.5 (15.47 $\mu\text{g}/\text{m}^3$) sits exactly at the WHO 24h guideline, meaning the average day in Cameroon is borderline unsafe. At a mean of 18.86 $\mu\text{g}/\text{m}^3$, every region exceeds the WHO annual limit of 5 $\mu\text{g}/\text{m}^3$.

Modelling decision: Log-transform PM2.5, PM10, Dust, and AOD before training. Use raw scale for CO, NO₂, O₃, SO₂ which show more symmetric distributions.

4.2 The Two Pollution Regimes

Regional analysis revealed that Cameroon does not have one air quality problem. It has two, with entirely different physical drivers.

Regime	Regions	Characteristics
Northern Sahel	Far North, North, Adamawa	Saharan dust transport via Harmattan winds. Mean PM2.5 24–29 $\mu\text{g}/\text{m}^3$ (1.6–1.9 $\times$ WHO). Episodic and extreme: Far North reaches $\sim 55 \mu\text{g}/\text{m}^3$ in February. Calendar-driven: strictly Nov–Feb.
Highland Burning	West, North West	Agricultural biomass burning, trapped by highland topography. West stays above 15 $\mu\text{g}/\text{m}^3$ even in the wet season, unlike all other non-northern regions. West climbed from 15.9 $\mu\text{g}/\text{m}^3$ (2020) to 33.4 $\mu\text{g}/\text{m}^3$ (2025), more than doubling.
Urban Coastal	Littoral, Centre	PM2.5 at or near the WHO limit (14–16 $\mu\text{g}/\text{m}^3$). Douala and Yaoundé benefit from equatorial moisture but face growing traffic and industry pollution.
Southern Forest	South West, South	Cleanest zone. 10–15 $\mu\text{g}/\text{m}^3$ average. Coastal cities like Kribi and Limbe drop below 5 $\mu\text{g}/\text{m}^3$ in September, the only part of Cameroon that briefly reaches the WHO

		annual limit.
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The PM10/PM2.5 ratio further confirms the distinction: Far North averages ~2.5 (Saharan coarse dust dominates) while southern regions average ~1.4–1.5 (combustion fine particles dominate). The West anomaly, a non-northern region ranking 3rd nationally in mean PM2.5, is the most policy-relevant discovery in this EDA because it represents a worsening trend that is independent of Harmattan seasonality.

Modelling decisions: (1) The two-regime discovery motivated a Two-Stage Harmattan Model experiment (separate regressors per regime). (2) City and region label encoders are essential: the model must distinguish spatial baselines, not just meteorology.

4.3 Harmattan Deep Dive: The Most Dangerous Days

Harmattan days represent 9.8% of the dataset but dominate the health risk profile:

Metric	Value	Implication
Total Harmattan days	8,532 (9.8%)	Minority of data; risk of under-representation in training
Harmattan PM2.5 mean	29.37 $\mu\text{g}/\text{m}^3$	1.66× the non-Harmattan mean of 17.72 $\mu\text{g}/\text{m}^3$
Far North multiplier	1.68×	Far North peak: ~55 $\mu\text{g}/\text{m}^3$ in February (3.7× WHO limit)
North multiplier	1.73×	
Adamawa multiplier	1.74×	Adamawa shows the strongest Harmattan effect relative to baseline
Cross-compound effect	PM10, CO, SO ₂ , NO ₂ all elevated	O ₃ does not respond; independent photochemical drivers confirmed

Modelling decision: Harmattan rows are weighted 3× in XGBoost training. Despite representing only 9.8% of training data, these are exactly the days that trigger health alerts. Under-predicting on Harmattan days is a much more costly error than under-predicting on a typical day.

4.4 Weather–PM2.5 Correlations

The full feature correlation ranking, computed on real CAMS rows only to avoid proxy distortion, revealed a clear physical story:

Rank	Feature	r	Physical Meaning
1	nh3_proxy (engineered)	+0.572	Biomass burning calendar index;

			outperforms every raw meteorological variable
2	humidity	-0.551	Low Harmattan humidity keeps particles suspended. This is the strongest raw predictor.
3	weather_code	-0.551	Rain/cloud codes correlate with clean air; clear sky correlates with high PM2.5 (see 4.5)
4	is_no_rain	+0.537	No wet scavenging → particles accumulate
5	is_dry_season	+0.521	Seasonal calendar flag for dry season months {11,12,1,2,3}
6	is_dust_event	+0.516	Binary flag: CAMS dust PM2.5 > 20 µg/m³
7	temp_range	+0.510	Large daily temperature swing = dry, cloudless Harmattan day
8	evapotranspiration	+0.504	High ET = dry surface = active dust source
9	vpd_max (v2 feature)	+0.501	Vapour pressure deficit; atmospheric water vapour deficit controls particle suspension time
10	soil_moisture	-0.494	Dry soil = active dust emission source
12	vpd (v2 feature)	+0.484	VPD via Tetens equation; nearly as predictive as humidity
14	wind_gust (v2 feature)	+0.186	Episodic gusts mobilise dust above a threshold; tree models capture this nonlinearity well
n/a	blh_min (v2 feature)	-0.099	Daily minimum boundary layer height; a shallow minimum traps pollution near the surface

The overall picture is unambiguous: atmospheric dryness drives PM2.5 in Cameroon. Features 1 through 12 are all proxies for the same physical state: dry air, no rain, dry soil, hot sunny days, and the seasonal burning window. This coherence gives high confidence that the features are capturing real atmospheric physics rather than spurious correlations.

Modelling decisions: All features with $|r| > 0.10$ validated for inclusion. The vpd_x_dry_season interaction term (SHAP rank #1 globally in the trained model) was engineered specifically from this analysis. VPD alone is strong but its effect is regime-dependent.

4.5 Counter-Intuitive Finding: Clear Sky = Highest PM2.5

Weather code analysis produced the most surprising finding in the EDA:

Sky Condition	Mean PM2.5	Explanation
Clear sky (WMO code 0)	32.80 $\mu\text{g}/\text{m}^3$	Counter-intuitive: Harmattan dust transport occurs under clear, dry anticyclonic conditions. Clear sky in northern Cameroon signals Saharan dust, not clean air.
Mainly clear / Partly cloudy	28.55 $\mu\text{g}/\text{m}^3$	Transition conditions: elevated but below the clear-sky peak
Rain / Showers	14.39 $\mu\text{g}/\text{m}^3$	Wet scavenging removes particles. Rain days are the cleanest by far, validating is_no_rain as a top-10 feature.

This finding has a direct implication for health communications: standard weather forecasts that show clear sky icons can inadvertently create false reassurance. The AirSense advisory system therefore always leads with PM2.5 rather than sky condition, and the Harmattan detection algorithm specifically fires on clear-sky + NE wind conditions rather than cloudy conditions.

Modelling decision: weather_code_enc is included as a feature. The non-linear, counter-intuitive relationship between sky condition and PM2.5 is exactly the kind of pattern tree-based models handle well. A linear model would likely assign the wrong sign to this feature.

4.6 Year-over-Year Trend: 2024 Was the Worst Year on Record

Analysing annual means by region revealed the most important public health finding in the entire project:

Year	National Mean PM2.5	Standard Deviation	Notable
2020	17.01 $\mu\text{g}/\text{m}^3$	5.23	Baseline year with moderate variability
2021	17.08 $\mu\text{g}/\text{m}^3$	5.17	Stable and comparable to 2020
2022	16.20 $\mu\text{g}/\text{m}^3$	6.64	Slight improvement; variability begins increasing
2023	18.74 $\mu\text{g}/\text{m}^3$	13.55	Sudden jump; variability more than doubles
2024	22.93 $\mu\text{g}/\text{m}^3$	19.19	Worst year on record. Far North peaked at 33.2 $\mu\text{g}/\text{m}^3$ (2.2× WHO)
2025	21.25 $\mu\text{g}/\text{m}^3$	15.48	Slightly improved from the 2024 peak but well above the 2020 baseline

The rising standard deviation (5.2 to 19.2) is particularly alarming. Not only is the mean PM2.5 worsening, but the year-to-year swings are becoming more extreme. Harmattan seasons are growing

more intense and more variable. The West region tells the starkest regional story: it climbed from ~15.9 $\mu\text{g}/\text{m}^3$ (2020) to 33.4 $\mu\text{g}/\text{m}^3$ (2025), more than doubling in five years, while not being a northern dust-affected region.

Modelling decision: The year feature is retained in the model. The EDA confirms it captures a genuine secular trend driven by climate change and expanding agricultural burning, not a data artefact from the proxy/CAMS boundary. A model trained without year would systematically underpredict 2024–2025 conditions.

4.7 Proxy Data Validation

Before using proxy-derived PM2.5 rows in model training, we validated two things: (1) whether the proxy distribution is close enough to real CAMS to be useful, and (2) whether including proxy rows distorts the feature correlations we rely on for feature selection.

Dataset	Rows	Mean PM2.5	Std Dev	Key Finding
Real CAMS measurements	49,400 (56.6%)	20.17 $\mu\text{g}/\text{m}^3$	15.71	Full physical range; reproduces extreme Harmattan events
Proxy targets (pre-CAMS)	37,840 (43.4%)	17.14 $\mu\text{g}/\text{m}^3$	5.24	Compressed distribution; the linear formula cannot reproduce extremes

The critical difference is standard deviation: proxy std = 5.24 vs CAMS std = 15.71, a 3x compression. The proxy rows cannot reproduce the extreme Harmattan episodes that are most important for health alerts. However, the means are similar ($\Delta = 3.03 \mu\text{g}/\text{m}^3$), so proxy rows are useful for capturing moderate-pollution seasonal patterns.

For correlation stability, we ran the full feature correlation ranking on CAMS-only rows. No feature showed $|\Delta r| > 0.05$ compared to the full dataset:

Feature	Full dataset r	CAMS-only r	Δ
humidity	-0.551	-0.567	-0.016
is_no_rain	+0.537	+0.528	-0.009
temp_range	+0.510	+0.525	+0.015
vpd	+0.484	+0.479	-0.005

Modelling decision: Train two models: an XGBoost mixed model (proxy + CAMS rows) and an XGBoost CAMS-only model, then compare both. The mixed model achieves slightly better overall MAE (6.268 vs 6.345) because the proxy rows add useful seasonal signal for normal conditions. The CAMS-only model is the conceptually cleaner result. Both are reported.

4.8 Summary: EDA Findings → Modelling Decisions

Every modelling choice in notebook 02 maps directly to an EDA finding:

EDA Finding	Modelling Decision	Where It Appears
PM2.5 skewness ≈ 2.37	Log-transform PM2.5, PM10, Dust, AOD	Notebook 02, Step 3: target preprocessing
Harmattan = 9.8% of data, 1.7 $\times$ PM2.5 multiplier	Harmattan rows weighted 3 $\times$ in training	XGBoost sample_weight parameter
nh3_proxy $r=+0.572$, top feature	Include nh3_proxy in feature set	features.json: feature slot #1 by importance
vpd_x_dry_season regime dependence (Sec 10b)	Engineer vpd_x_dry_season interaction feature	SHAP rank #1 globally in deployed model
Clear sky = highest PM2.5 (non-linear)	Include weather_code_enc; use tree model	XGBoost preferred over linear models
Two pollution regimes confirmed	Experiment with Two-Stage Harmattan Model	Notebook 02 Step 6b; single XGBoost ultimately wins
2024 secular trend confirmed by CAMS validation	Retain year as a feature	Top-5 SHAP feature across most regions
Proxy distribution compressed (std 5.24 vs 15.71)	Train CAMS-only and mixed models separately	Two deployment candidates compared in Sec 4 of this report
blh_min $r=-0.099$ cleaner than blh_mean $r=+0.125$	Use both BLH features; keep blh_min	features.json: both retained after ablation
Compound-specific meteorological sensitivity	Train one XGBoost model per compound	models_multi.pkl: 8 compound models
O ₃ does not respond to Harmattan	O ₃ modelled separately; no Harmattan weight	Separate model in models_multi.pkl

5. Comparative Model Analysis

All models were evaluated on the same held-out test split using real CAMS PM2.5 measurements as the target. Models trained on the synthetic proxy target (the starter formula) are marked separately and are not directly comparable to real-target models.

5.1 Full Model Comparison Table

Model	MAE	R ²	Role / Notes
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	($\mu\text{g}/\text{m}^3$)		
BASELINE: Starter Formula	9.047	0.208	Official hackathon baseline; targets synthetic proxy, not real PM2.5
BASELINE: City-Month Mean	7.521	0.454	Simple statistical baseline using city and month average PM2.5
XGBoost (CAMS-only, Optuna)	6.345	0.596	Primary nowcasting model. No circularity; real satellite targets only
XGBoost Mixed (FINAL DEPLOY)	6.268	0.603	★ Deployment model. CAMS + proxy data, Optuna-tuned, best overall balance
Two-Stage Harmattan Model	6.118	0.646	Separate regressors for Harmattan vs non-Harmattan regime. Single XGBoost wins in production.
LightGBM + XGBoost Ensemble	5.969	0.651	Stacked ensemble. No statistically significant gain over single model at deployment cost.
XGBoost + Lag Features (Operational)	0.685	0.992	When real prior PM2.5 is available; near-perfect but requires actual sensor readings
XGBoost Optuna CAMS-only (Nowcast)	5.936	n/a	Same-day nowcasting with full CAMS feature set; best meteorology-only MAE
Transformer Day+1 (Early Warning)	7.048	0.597	24-hour ahead forecast; temporal sequence model for a 72-hour early warning system
Transformer Day+3 (Early Warning)	7.260	0.569	72-hour ahead forecast; gradual performance degradation expected over the horizon
GraphSAGE GNN (Spatial)	8.186	0.291	Spatial interpolation to ungauged cities; propagates information across the city graph
Ridge Regression (Baseline)	7.521	0.454	City-month average fallback, used as a statistical comparison point
Multi-Output XGBoost (8 compounds)	5.93–39.1	–0.17–0.70	Separate models per compound. PM2.5 performs best (MAE 5.93); dust performs worst (negative R^2)

5.2 Alert Classifier Comparison

PM2.5 exceedance prediction (WHO 24h limit $> 15 \mu\text{g}/\text{m}^3$) was evaluated as a classification task. Three approaches were compared:

Classifier	AUC	F1	Notes
Platt-Calibrated Logistic (Deployed)	0.921	0.847	$\text{sigmoid}(0.2221 \times \text{PM2.5} - 3.0372)$ with per-city recalibration for distribution shift
Temperature Scaling (Selected)	0.921	0.857	Best calibration method; selected over Platt, isotonic, and no calibration

Rule-Based Threshold (PM2.5 > 15)	n/a	0.905	Higher F1 but no probability output; misses dangerous days near the boundary
XGBoost + Lag Features Classifier	n/a	Best (0.685 MAE)	Near-perfect when real prior readings are available (operational setting)

The rule-based threshold achieves higher F1 (0.905 vs 0.847), but the calibrated probabilistic classifier provides actionable alert probabilities and catches dangerous days near the boundary that a hard cutoff misses. For health advisory purposes, calibrated probabilities are more valuable than raw binary predictions.

5.3 Architecture-Level Comparison

Three complementary model architectures are deployed simultaneously, each serving a distinct operational role:

Architecture	Operational Role	MAE	When to Use
XGBoost (CAMS + meteo)	Real-time nowcasting	6.27 $\mu\text{g}/\text{m}^3$	Primary path when meteorological forecast features are available from Open-Meteo API
XGBoost + Lag Features	Operational (sensor-assisted)	0.69 $\mu\text{g}/\text{m}^3$	Used when IoT sensor readings from the previous day are available via the /ingest endpoint
Transformer (Day+1 to Day+3)	72-hour early warning	7.05–7.26 $\mu\text{g}/\text{m}^3$	Used for school and farm advisories requiring multi-day advance notice
GraphSAGE GNN	Spatial interpolation	8.19 $\mu\text{g}/\text{m}^3$	Used for generating predictions for cities outside the training set using spatial proximity

6. Key Technical Innovations

The project introduces ten original contributions beyond the standard hackathon scope:

#	Innovation	Description
1	Breath of Fresh Air Index (BFAI)	Composite 0 to 100 score combining PM2.5 (50%), wind speed (25%), and humidity (25%). More intuitive than raw $\mu\text{g}/\text{m}^3$ and used directly in school and farm advisories

2	Harmattan Early-Warning System	Detects Saharan dust intrusions using NE wind bearing, dry-season flag, and PM2.5 spike. Triggers 24-hour advance advisories in northern regions
3	Conformal Prediction Intervals	Distribution-free 90% confidence bounds per city, achieving 97.7% coverage. No Gaussian assumptions required, making it appropriate for skewed PM2.5 distributions
4	Platt-Calibrated Alert Classifier	Logistic regression maps XGBoost output to exceedance probability (F1=0.847, AUC=0.921). Per-city recalibration corrects for regional distribution shift
5	Dynamic Source Attribution	5-source decomposition covering Dust, Biomass, Traffic, Industry, and Secondary Aerosol. Computed from live meteorological conditions and SHAP weights; updates per city and month
6	CMIP6 Climate Projections	SSP2-4.5 and SSP5-8.5 scenarios projected to 2050 with bias correction from the historical CAMS baseline. Maroua is projected at -15% reduction under SSP2
7	Conversational AI Assistant	Groq Llama 3.1 8B on the free tier (14k req/day) with live PM2.5 context, forecasts, and SHAP data injected into the system prompt. Supports streaming responses, bilingual EN/FR, and a rule-based fallback when no API key is present
8	One-Click PDF Report Export	Per-city downloadable PDF covering current status, 7-day forecast table, source attribution, health impact, and recommendations. Fully bilingual via ReportLab
9	API-First with Silent Fallback	Dashboard routes all predictions through the FastAPI backend. If the backend is unreachable, the local XGBoost model loads silently with no visible failure to the user
10	Configurable Reference Standards	WHO 2021 (15 $\mu\text{g}/\text{m}^3$), EU 2024 (25), US EPA (35), ECOWAS (50), or a custom threshold. All visualisations and alert counts update dynamically on selection
11	Progressive Web App (PWA)	Full offline-capable Progressive Web App built on top of Streamlit. The service worker (sw.js) uses cache-first for assets and network-first for navigation, with an offline fallback page. The Web App Manifest sets display to standalone and includes 192 and 512px icons. Installable on Android and iOS via Add to Home Screen. Designed to work on both local dev and Streamlit Cloud paths without any config changes.

6.2 City Coverage: 71 Cities Across All 10 Regions

The dashboard covers 71 Cameroonian cities, distributed across all 10 administrative regions. The breakdown below shows the current coverage and clarifies that the 40-city official dataset was expanded to 71 for the dashboard deployment:

Region	Cities	City Names
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West	11	Babadjou, Bafang, Bafoussam, Baham, Bandjoun, Bangangté, Bazou, Dschang, Foumban, Foubot, Mbouda
Centre	10	Afanloum, Akono, Akonolinga, Ayos, Bafia, Eséka, Mbalmayo, Mbandjock, Monatélé, Yaoundé
North West	8	Babessi, Bafut, Bakou, Bali, Bamenda, Fundong, Kumbo, Mbengwi
Littoral	7	Dizangué, Douala, Edéa, Loum, Manjo, Mbanga, Melong
East	7	Abong-Mbang, Batouri, Bélabo, Bertoua, Bétaré-Oya, Doumé, Garoua-Boulai
South West	7	Akwaya, Bangem, Buéa, Fontem, Kumba, Limbé, Mamfé
South	7	Akoéman, Akom II, Ambam, Campo, Ebolowa, Kribi, Lolodorf
Far North	6	Bogo, Kaélé, Kousséri, Maroua, Mokolo, Mora
North	5	Bibémi, Figuil, Garoua, Guider, Lagdo
Adamawa	3	Banyo, Meiganga, Ngaoundéré
TOTAL	71	Full national coverage across all 10 administrative regions

7. Reproducibility Guide: Step by Step

This section provides a complete, ordered path to reproduce every result in this project from scratch. All steps are designed to be run in sequence. Estimated total time: 2–4 hours (notebook execution is the longest step).

Step 0: Prerequisites

Ensure the following are installed on your system before proceeding:

- Python 3.10 or higher (3.11 recommended)
- Git (for cloning the repository)
- Jupyter (pip install jupyter) for running notebooks
- Optional: Free Groq API key from console.groq.com for the AI assistant
- Optional: Docker and Docker Compose for containerised deployment

Step 1: Clone and Install

Clone the repository and install all Python dependencies:

```
git clone https://github.com/dze-shalom/airsense_indabax_hackathon_2026
cd airsense_indabax_hackathon_2026
pip install -r requirements.txt
```

The requirements.txt pins all package versions. Key dependencies and their roles:

Package	Version	Role
xgboost	>=2.0.0	Primary prediction model. Version 2.x or higher is required for native pred_contribs=True SHAP support
streamlit	>=1.32.0	Dashboard frontend; version 1.32 or higher is required for theme detection and custom CSS injection
fastapi + uvicorn	>=0.110.0	REST API backend; uvicorn standard extras include websocket and http/2 support
shap	>=0.44.0	SHAP feature attribution; version must match XGBoost to avoid TreeExplainer incompatibility
optuna	>=3.4.0	Hyperparameter optimisation for XGBoost tuning in notebook 02
lightgbm	>=4.0.0	Ensemble component used in the LightGBM and XGBoost stack evaluation
reportlab	>=4.0.0	PDF report generation for bilingual city-level exports from the dashboard Science tab

Step 2: Prepare the Official Dataset

The hackathon provides the base dataset as an Excel file. Before running notebooks, convert it to CSV:

```
# Open Dataset_complet_Meteo.xlsx in Excel or LibreOffice
# File → Save As → CSV UTF-8 (important: avoids date auto-formatting)
# Save to: notebooks/data/Dataset_complet_Meteo.csv
```

Why CSV and not Excel directly: Excel silently auto-formats integer date codes as human-readable dates, which breaks date parsing in pandas. The CSV export preserves raw values.

Step 3: Run Notebooks in Order

Four notebooks must be executed in the following sequence. Each notebook reads the parquet file produced by the previous one.

Notebook 00: Data Cleaning & Feature Engineering

Input: Dataset_complet_Meteo.csv | Output: data/airsense_eda_ready.parquet (87,240 × 80, 0 nulls)

1. Fetches 12 weather variables from Open-Meteo Archive API (humidity, VPD, BLH, etc.)
2. Fetches 8 CAMS compounds from Open-Meteo Air Quality API (PM2.5, PM10, dust, CO, NO₂, O₃, SO₂, AOD)
3. Builds and validates proxy targets for the 2020–2022 pre-CAMS period

4. Saves the final null-free parquet file

Notebook 01: Exploratory Data Analysis

Input: airsense_eda_ready.parquet | Output: Charts (embedded in notebook, no files saved)

5. Analyses target distributions, regional patterns, Harmattan seasonality, and feature correlations
6. Documents the two pollution regimes (northern Sahel vs highland biomass burning)
7. Produces the 12 modelling decisions that drive notebook 02

Notebook 02: Model Training & Evaluation

Input: airsense_eda_ready.parquet | Output: 20 model artefacts saved to models/

8. Trains XGBoost (CAMS-only and mixed) with Optuna hyperparameter optimisation
9. Computes conformal prediction intervals ($q_{\text{hat}} = \pm 17.34 \mu\text{g}/\text{m}^3$, 97.7% coverage)
10. Trains Platt-calibrated alert classifier (AUC=0.921, F1=0.847)
11. Computes per-region SHAP values and saves region_shap.pkl
12. Trains Transformer (Day+1/+2/+3) and GraphSAGE GNN
13. Generates CMIP6 2050 projections under SSP2-4.5 and SSP5-8.5
14. Saves all 20 artefacts to the models/ directory

Notebook 03: Pitch Charts

Input: airsense_eda_ready.parquet + models/ | Output: 9 PNG charts in outputs/

15. Generates publication-ready charts for the pitch deck and demo video

```
# Run all notebooks non-interactively:
jupyter nbconvert --to notebook --execute notebooks/00_Data_Cleaning.ipynb
jupyter nbconvert --to notebook --execute notebooks/01_EDA.ipynb
jupyter nbconvert --to notebook --execute notebooks/02_Modelling.ipynb
jupyter nbconvert --to notebook --execute notebooks/03_Pitch_Charts.ipynb
```

Step 4: Rebuild Model Artefacts (if Skipping Notebooks)

If you do not run the full notebooks (e.g. no API access to CAMS), run the setup script to generate functional stub artefacts that allow the dashboard to load without crashing:

```
python scripts/setup_models.py
```

This script (~5 seconds) rebuilds all .pkl artefacts that are tracked via Git LFS. The main prediction model (xgb_pm25.json, 1.4 MB) is committed directly to Git and does not require LFS. It is always available after cloning.

Step 5: Configure Secrets (Optional but Recommended)

Create the secrets file for API keys. This file is gitignored. Never commit it.

```
mkdir .streamlit
# Create .streamlit/secrets.toml:
GROQ_API_KEY = "gsk_..." # Free at console.groq.com
AIRSENSE_API_URL = "http://localhost:8000" # Optional API backend
```

Without the Groq key, the dashboard still works fully. The AI assistant falls back to rule-based responses and the policy brief uses data-driven templates. The AIRSENSE_API_URL is only needed for API-first mode.

Step 6: Launch the Dashboard

```
cd dashboard
streamlit run app.py
```

Open <http://localhost:8501> in your browser. The dashboard loads in under 10 seconds on first run (model loading). Subsequent page navigations are faster as models are cached in Streamlit session state.

Step 7: Launch the API (Optional)

The dashboard works without the API, using local model fallback. To enable API-first mode with full endpoint access:

```
uvicorn api.main:app --reload --port 8000
```

Interactive API documentation is available at <http://localhost:8000/docs> (Swagger UI). All 20+ endpoints are documented with request/response schemas.

Step 8: One-Shot Setup (Shortcut)

For a single-command setup that runs all steps and launches both services:

```
bash setup.sh
```

The setup.sh script installs requirements, runs feature engineering, trains the XGBoost model, trains the REINFORCE RL threshold agent, and launches both the API (background) and dashboard (foreground).

Step 9: Deploy to Render.com

Production deployment requires pushing to a GitHub repository connected to Render.com. The `render.yaml` configuration file defines two services (dashboard + API) and is already included in the repository root.

16. Push the repository to GitHub
17. Connect the GitHub repo to Render.com
18. Set `GROQ_API_KEY` and `AIRSENSE_API_URL` in the Render dashboard environment variables
19. Render auto-deploys on every git push to main

The live deployment is accessible at: <https://airsensecameroon.streamlit.app>

Step 10: Adding a New City

The system is designed so that adding a city requires changes in exactly two files. No retraining is needed. The XGBoost model generalises spatially to any city via the region-level label encoder, and Open-Meteo provides live weather for any global lat/lon coordinate.

Where cities are defined

All 71 cities live in a single `CITIES` dictionary in `dashboard/config.py`. Each entry is a (name, lat, lon) tuple grouped by region:

```
# dashboard/config.py: CITIES dictionary
CITIES = {
    "Littoral": [
        ("Douala", 4.0483, 9.7043),
        ("NewCity", LAT, LON), # ← add your city here
    ],
}
```

What happens automatically after adding

Feature	Behaviour for New City
7-day PM2.5 forecast	Works immediately. Open-Meteo fetches live weather for any lat/lon worldwide and XGBoost uses the region-level encoder for spatial identity.
Conformal prediction intervals	Applied from the region-level <code>q_hat</code> in <code>conformal_intervals.json</code> ; no per-city calibration is needed.
SHAP feature attribution	Uses the region's pre-computed SHAP fingerprint from <code>region_shap.pkl</code> . Per-city SHAP values are only available for the 40 originally trained cities.
Alert probability	Platt calibration uses a global sigmoid formula and works for any city whose PM2.5 falls within the trained range.
Overview map & rankings	The new city appears automatically on the national choropleth map and city rankings on the next page load.

Bilingual advisories	All EN/FR advisory text is region-based rather than city-specific, so it works immediately for any new city in an existing region.
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Accuracy expectations for new cities

Spatial generalisation was validated in the modelling notebook. For cities within Cameroon but outside the 40 training cities (L1 level), the model achieves MAE = 3.72–6.62 $\mu\text{g}/\text{m}^3$ and $R^2 = 0.51$ –0.68, comparable to the trained-city performance. For cross-border cities in neighbouring countries (L2), accuracy drops (N'Djamena MAE = 12.90, $R^2 = 0.09$). For fully out-of-domain cities (L3, e.g. Nairobi), the model still produces predictions (MAE = 7.10, $R^2 = 0.51$) because the meteorological drivers of PM2.5 are universal, but caution is warranted.

Optional: add to setup_models.py for full stub support

If you also want the stub artefacts (label_encoders.pkl, city_calibrators.pkl) to include the new city, add the same tuple to the CITIES dictionary in scripts/setup_models.py and re-run:

```
python scripts/setup_models.py
```

This step is optional. The dashboard works without it, falling back to region-level encoders for any city not in the pkl files.

8. Known Issues & Reproducibility Gotchas

The following issues were encountered during development and are documented here to prevent wasted time during reproduction:

Issue	Root Cause	Resolution
SHAP / XGBoost version mismatch	TreeExplainer in shap versions below 0.44 does not support the XGBoost 2.x booster format	Use native XGBoost SHAP: <code>model.predict(dmatrix, pred_contribs=True)</code> . Do not use <code>shap.TreeExplainer</code> on XGBoost 2.x models
Theme-adaptive CSS failure	Hardcoded dark-theme colour variables in inline style attributes override Streamlit theme switching	All CSS colour values must use <code>st.session_state['theme']</code> conditionally. Never hardcode dark colours in inline styles that apply to both themes
Open-Meteo API variable name error	Open-Meteo silently ignores unrecognised variable names, returning an empty column with no error raised	Always verify variable names against the Open-Meteo documentation before adding to the fetch list. Check for None columns after fetching
Proxy data target leakage	Using proxy PM2.5 as the training target creates a model that inverts the proxy formula rather than learning atmospheric physics	Use real CAMS rows only for training targets. Proxy rows are acceptable as additional feature rows if the target column is excluded from proxy rows

Excel date auto-formatting	Excel auto-converts integer date codes such as 45000 to human-readable dates when opening xlsx files	Always save the official .xlsx as CSV UTF-8 before loading into pandas. Do not use <code>pd.read_excel()</code> directly on the hackathon dataset
Git LFS / .pkl missing on clone	Several .pkl files are tracked via Git LFS. Cloning without LFS enabled produces pointer files rather than the actual model objects	Run python scripts/setup_models.py immediately after cloning to regenerate all .pkl stub files. xgb_pm25.json is not LFS-tracked and is always available
Dust / O₃ proxy negative R²	Engineered proxy formulas for dust_target and o3_target produced negative R ² against real CAMS values, meaning they performed worse than a simple mean predictor	Replace negative-R ² proxies with city-month medians computed from real CAMS rows. Always validate proxies before applying to the full dataset

9. API Reference Summary

The FastAPI backend exposes 20+ REST endpoints. Below are the most important ones for reproducing dashboard functionality:

Method	Endpoint	Description
GET	/health	Service health check; used by the dashboard to decide between API and local fallback
GET	/forecast/{city}	7-day XGBoost forecast powered by Open-Meteo data, with conformal confidence intervals
GET	/alert-prob/{city}	Platt-calibrated probability that PM2.5 will exceed the WHO 24h limit in the next 24 hours
GET	/explain/{city}	Region-specific SHAP fingerprint showing the top 7 feature importances for the city
GET	/climate/{city}	CMIP6 PM2.5 projections to 2050 under SSP2-4.5 and SSP5-8.5 scenarios
GET	/bfai/{city}	Breath of Fresh Air Index, the 0 to 100 composite score
GET	/school-advisory/{city}	School outdoor action level from 1 to 4, based on current PM2.5 and BFAI
GET	/africa-benchmark	Cameroon PM2.5 compared against 12 African cities for contextual benchmarking
POST	/ingest	ESP32 IoT sensor data ingestion; enables the lag-feature operational model

10. Team & Data Sources

9.1 Team

Member	Role
Dze-Kum Shalom Chow	Lead developer: ML modelling, FastAPI backend, Streamlit dashboard design, and all 11 innovations
Ayanda Blessing Khumalo	Statistical lead: statistical validation, hypothesis testing, and model evaluation framework
MALLA NDASSI Marie Ange	Domain expert: atmospheric physics insights, feature justification, and Harmattan regime analysis
Mih Ndum Lizette	Data pipeline and ETL engineering: CAMS and Open-Meteo data fetching, proxy target validation

9.2 Data Sources

- CAMS Reanalysis (Copernicus Atmosphere Monitoring Service / ECMWF): satellite-measured PM2.5 and 7 additional compounds covering 2022 to 2025
- Open-Meteo ERA5 Archive API: historical meteorological variables including BLH, VPD, precipitation, wind, temperature, and solar radiation from 2020 to 2025
- Open-Meteo Air Quality API: additional CAMS compounds accessed through Open-Meteo's unified interface
- CMIP6 (Coupled Model Intercomparison Project Phase 6): SSP2-4.5 and SSP5-8.5 climate scenario data for 2050 projections
- WHO Global Air Quality Guidelines 2021: reference standard for the 24h PM2.5 limit of 15 $\mu\text{g}/\text{m}^3$
- Official IndabaX hackathon dataset (Dataset_complet_Meteo): 87,240 rows covering 40 cities and 26 meteorological variables